

SV Students Recommend

QA WORKBOOK · SRS

SPECIFICATION

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FOR STUDENTS

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YOUR TASK

API test bank (pick 6/10)

UI test bank (pick 14/24)

What to submit

SOFTWARE REQUIREMENTS SPECIFICATION

SV Students *Recommend*

QA Testing Project – API & UI End-to-End testing with Playwright (Python)

APP <https://sv-students-recommend.onrender.com>

1. Introduction

This document defines the requirements for the “SV Students Recommend” web application. Students can log in, share recommendations, comment and rate them, buy SV College merchandise, and manage their profiles. Admins moderate the platform.

This SRS is the single source of truth for your testing project. Read it fully before you start. At the end you will find a **test bank**: choose **6 of 10** API tests and **14 of 24** UI tests to build and run.

How to use this page

Sections 1–5 describe the system. Section 6 shows how to store your project on GitHub. Sections 7–8 help you write tests in Playwright + Python. The test bank lets you pick which tests to implement – the live counters track your selection.

1.1 Glossary

Term	Meaning
SRS	Software Requirements Specification – this document.
UI	User Interface – what the user sees and clicks in the browser.
E2E	End-to-End – a test that follows a full user journey.
API	The server endpoints the website talks to.
RBAC	Role-Based Access Control – Admin vs. regular-user permissions.
Token	A secret string returned after login, sent on protected API calls.
Positive / Negative / Boundary	Valid input / wrong input / edge values (e.g. exactly 6 chars).

2. User Roles

2.1 Regular User

Can register, log in, view recommendations, add recommendations, add comments and ratings, edit/delete their own recommendations, manage their own profile, and buy items from the store.

2.2 Admin User

Has all regular-user permissions, **plus**:

- Open the System page (`/pages/admin.html`).
- Delete **any** comment.
- Delete **any** recommendation by ID.
- Ban / unban users (a banned user cannot log in).
- Add / remove emails from the blacklist (blacklisted emails cannot register or log in).
- Suspend new recommendations for everyone.

Test accounts

Admin: `admin@svcollege.co.il` **Password:** `test1234`

Tip: create your own regular user via Register, using a unique email each run, e.g. `test_<random>@example.com` .

3. Functional Requirements

3.1 Authentication & Registration

3.1.1 Login – email + password, Sign In. Eye icon shows/hides the password. “Continue with Google” button. “Forgot password?” link. Wrong credentials show a clear error. Banned users and blacklisted emails cannot log in.

3.1.2 Registration – Student Name, Email, Password (minimum 6 English characters). “Continue with Google” available.

3.1.3 Forgot Password – popup; enter email and send a reset link.

3.2 Toolbars

3.2.1 Upper toolbar: Logo & Home → Home; Help/About → popups; Store; My Profile; System (admin only); Cart (item count); Logout → Login; + Add Recommendation → form.

3.2.2 Footer: “© 2026 All rights reserved”, SV College link, Accessibility link.

3.3 Recommendations

3.3.1 Home: list of cards; filter by All / Book / Movie / Series / Activity / Other. **3.3.2 Detail:** image + details; owner can edit/delete; any user can comment with a 1–5 rating; admin deletes any comment. **3.3.3 Add:** mandatory = Category, Name, Your Name; optional = Description, Website, Image; empty mandatory field blocks submit.

3.4 Profile & Admin System

3.4.1 Profile: change password, delete account, logout. **3.4.2 System (admin):** ban user, email blacklist, suspend new recommendations, delete recommendation by UUID.

3.5 Store & Checkout

3.5.1 Store: T-Shirt (50 NIS), Cup (20 NIS), Add to cart. **3.5.2 Cart:** change quantity, recalculated totals, remove item, proceed to payment. **3.5.3 Payment:** order summary + mandatory Full Name, Address, Credit Card Number, CVV, Expiration Date; Place Order; empty mandatory fields block submission.

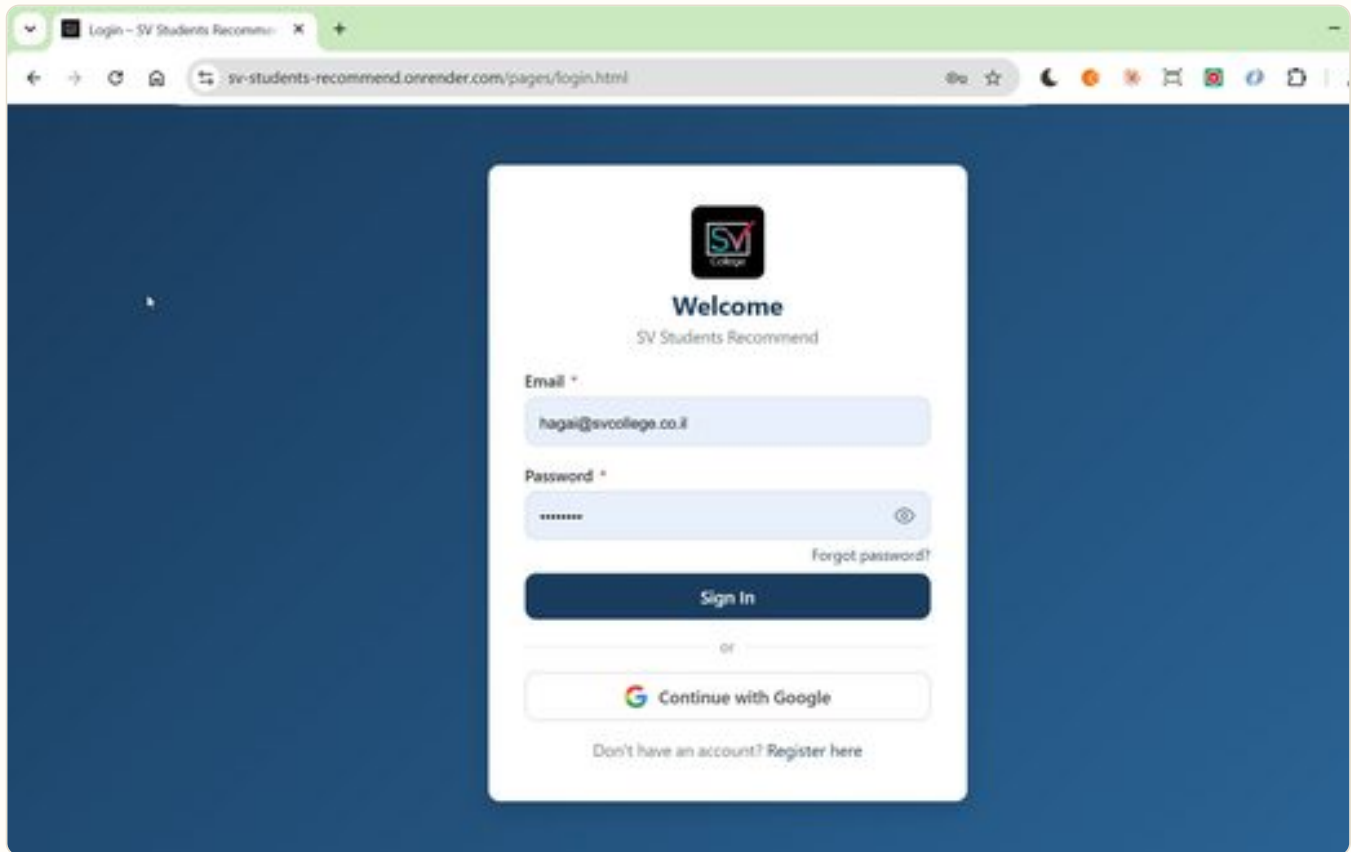
4. Page Reference

Screenshots of every page with its URL, so you know what you are testing and where to navigate.

4.1 Login Page

URL <https://sv-students-recommend.onrender.com/pages/login.html>

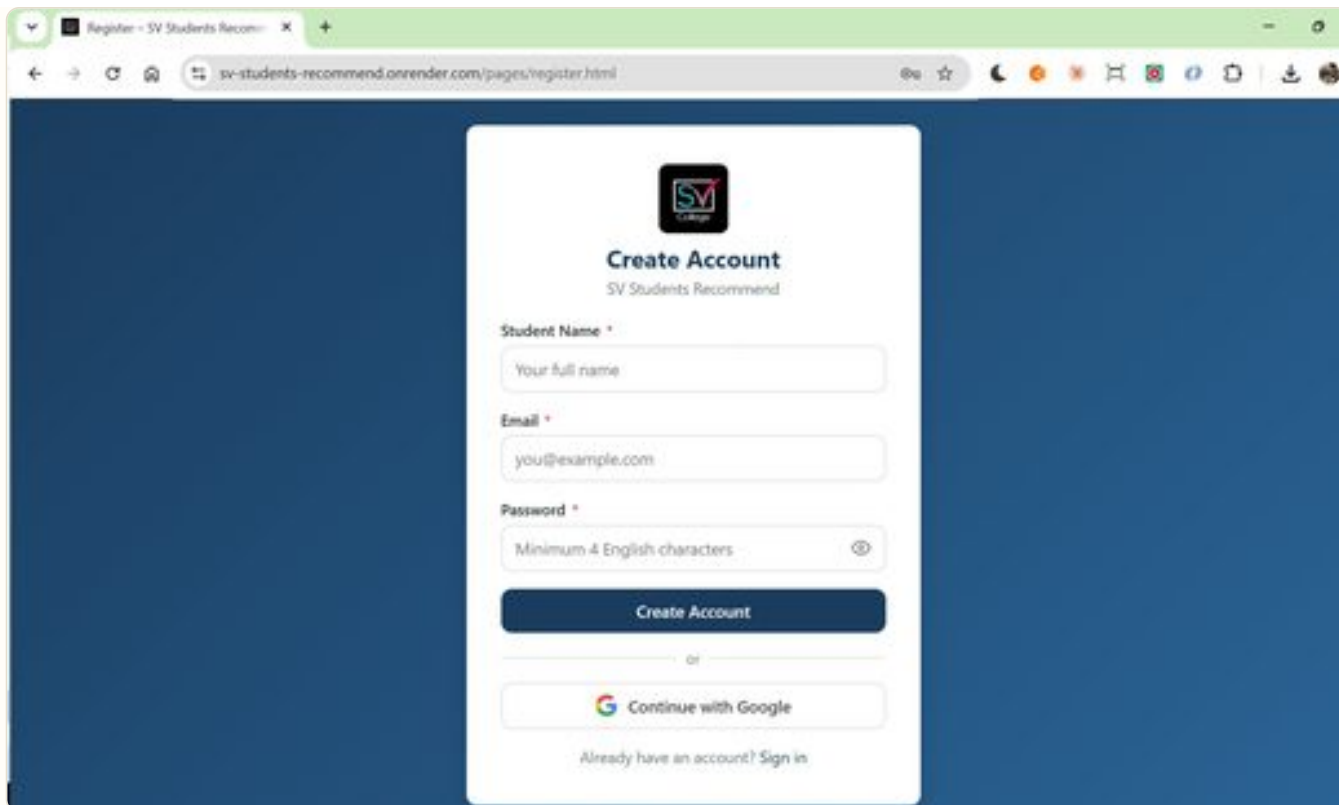
Note the password eye icon, “Continue with Google”, and “Forgot password?”.



4.2 Registration Page

URL <https://sv-students-recommend.onrender.com/pages/register.html>

Password hint: “Minimum 6 English characters” – drives your boundary tests.



Register - SV Students Recommend

sv-students-recommend.onrender.com/pages/register.html

Create Account
SV Students Recommend

Student Name *

Your full name

Email *

you@example.com

Password *

Minimum 4 English characters

Create Account

or

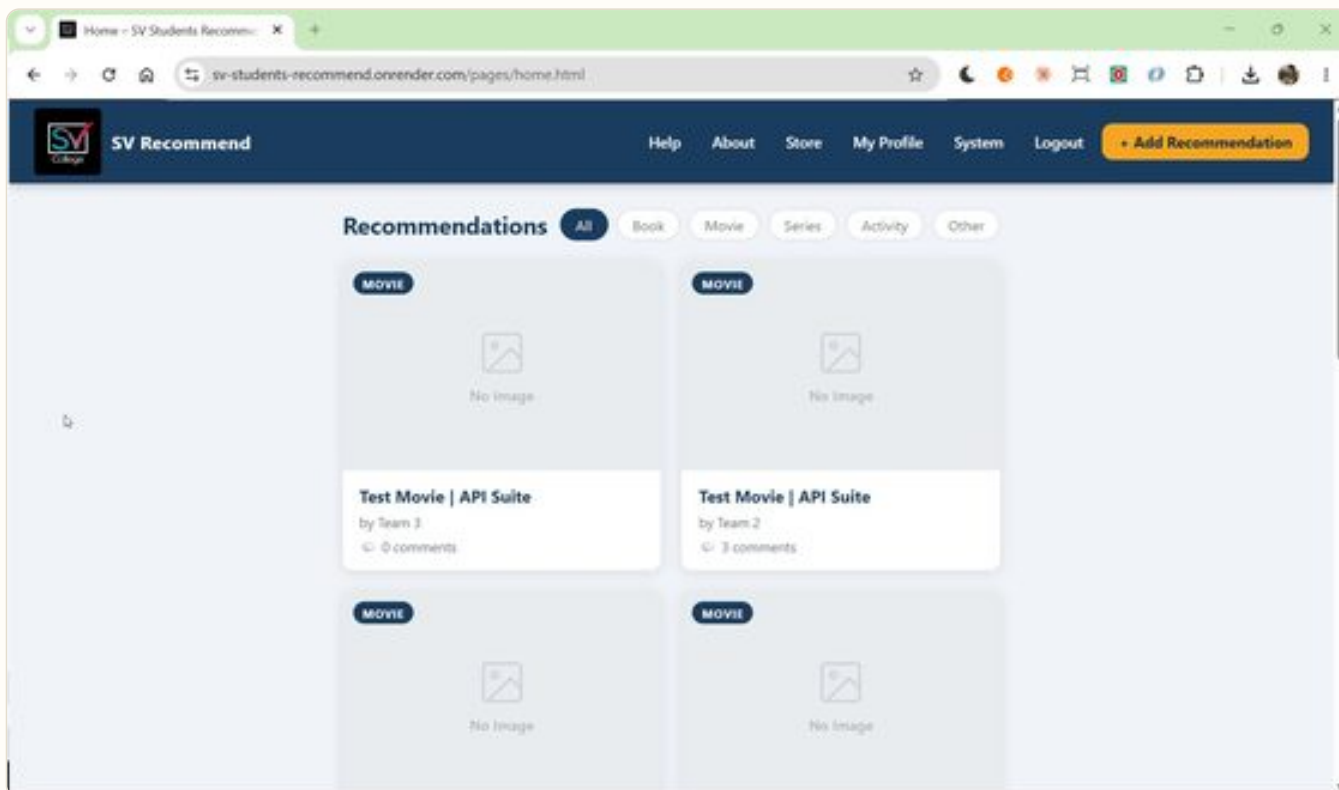
Continue with Google

Already have an account? Sign in

4.3 Home Page

URL <https://sv-students-recommend.onrender.com/pages/home.html>

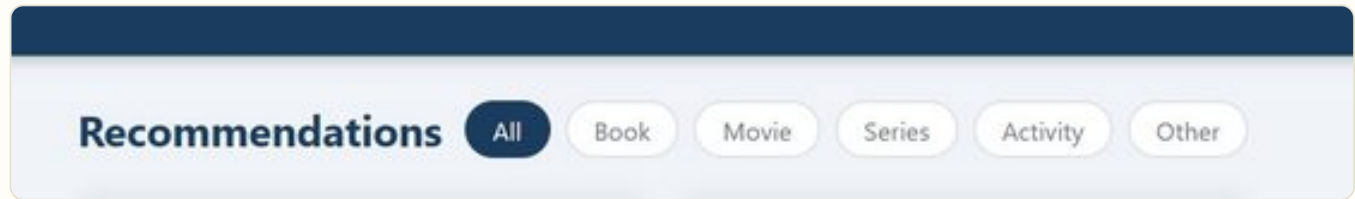
Cards with category, title, author, comment count.



4.3b Category Filter

URL <https://sv-students-recommend.onrender.com/pages/home.html>

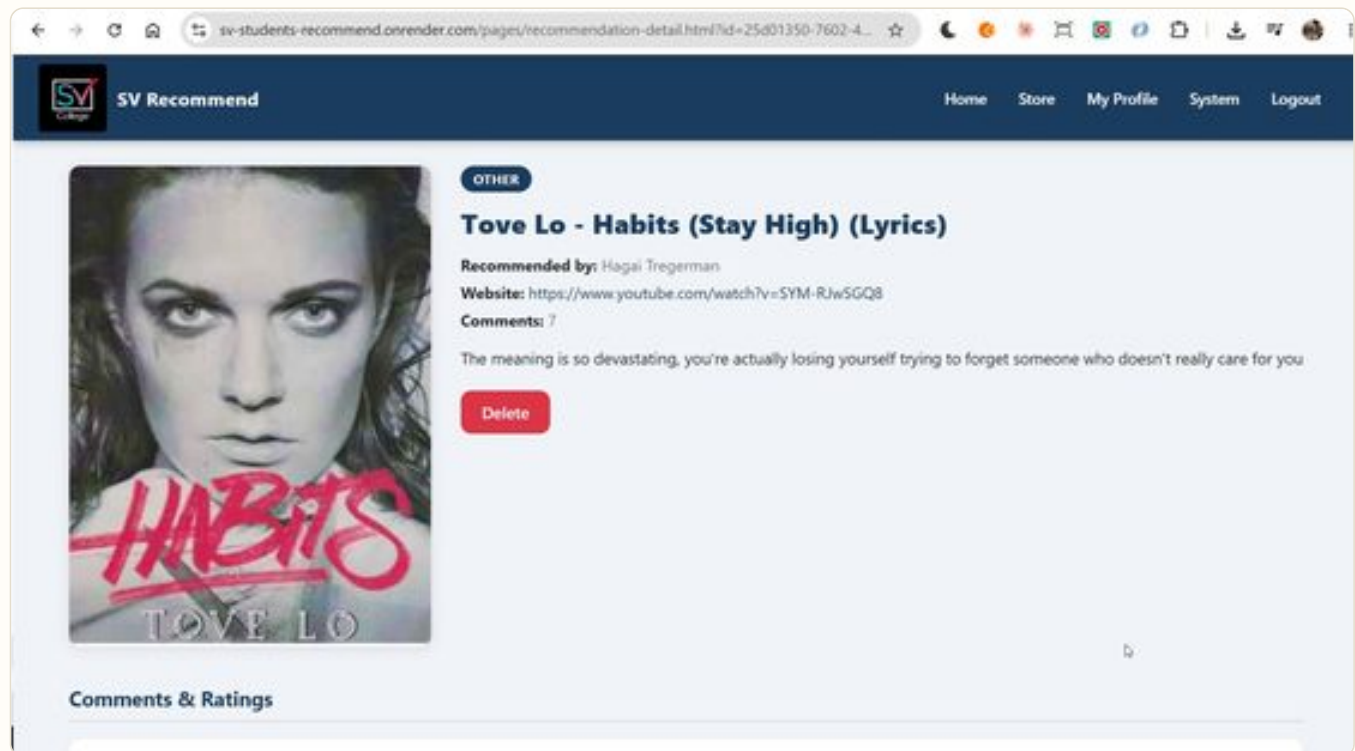
All / Book / Movie / Series / Activity / Other.



4.4 Recommendation Detail

URL <https://sv-students-recommend.onrender.com/pages/recommendation-detail.html>

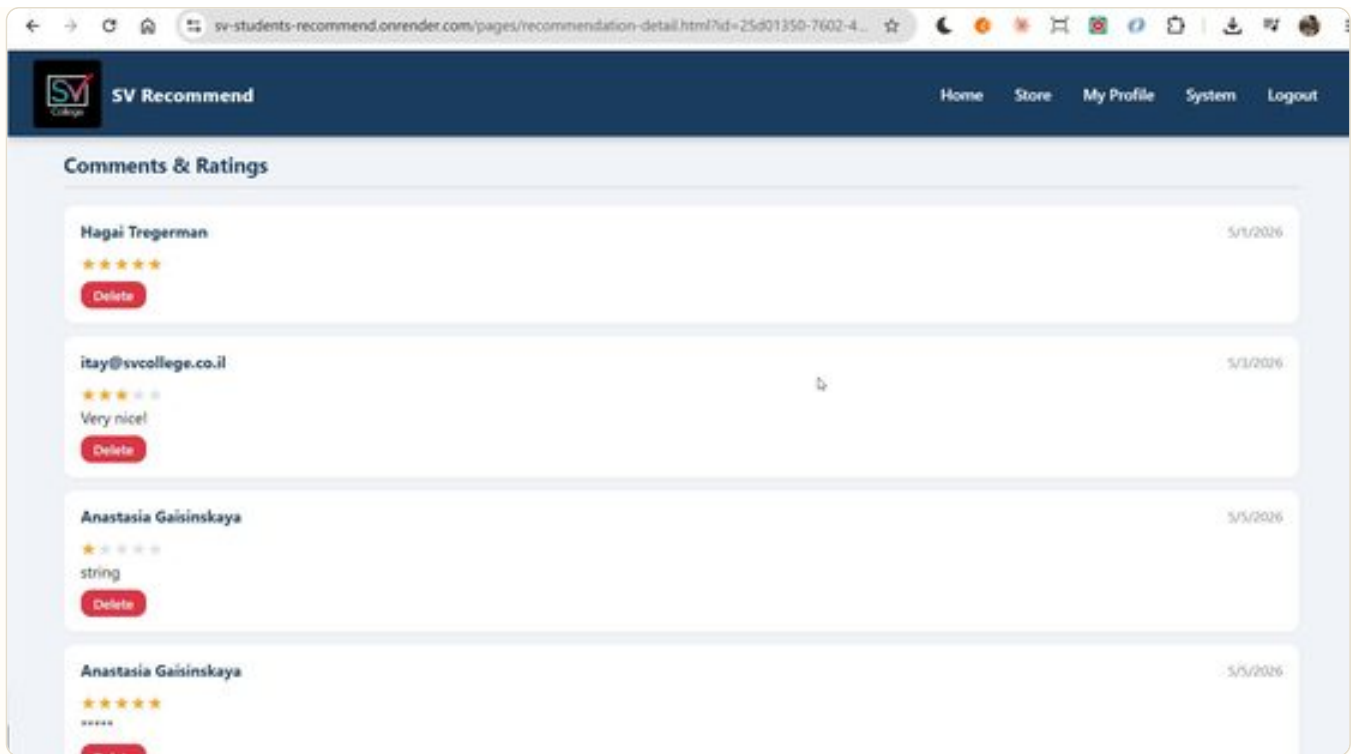
Image, details, website link, Delete (owner or admin).



4.4b Comments & Ratings

URL <https://sv-students-recommend.onrender.com/pages/recommendation-detail.html>

Admin sees Delete on every comment.



4.5 Add Recommendation

URL <https://sv-students-recommend.onrender.com/pages/add-recommendation.html>

Mandatory: Category, Name, Your Name. Optional: Description, Website, Image.

The screenshot displays the 'Add Recommendation' form on the SV Recommend website. The form is titled 'Add Recommendation' and contains several input fields. The 'Category' dropdown is set to 'Movie'. The 'Recommendation Name' field contains the text 'e.g. Inception, The Alchemist...'. The 'Your Name' field contains the text 'Hagai'. The 'Description' field contains the text 'Why do you recommend this? (optional)'. The 'Website Link' field contains the text 'https://...'. The 'Image' field is empty.

Add Recommendation

Category *
Movie

Recommendation Name *
e.g. Inception, The Alchemist...

Your Name *
Hagai

Description
Why do you recommend this? (optional)

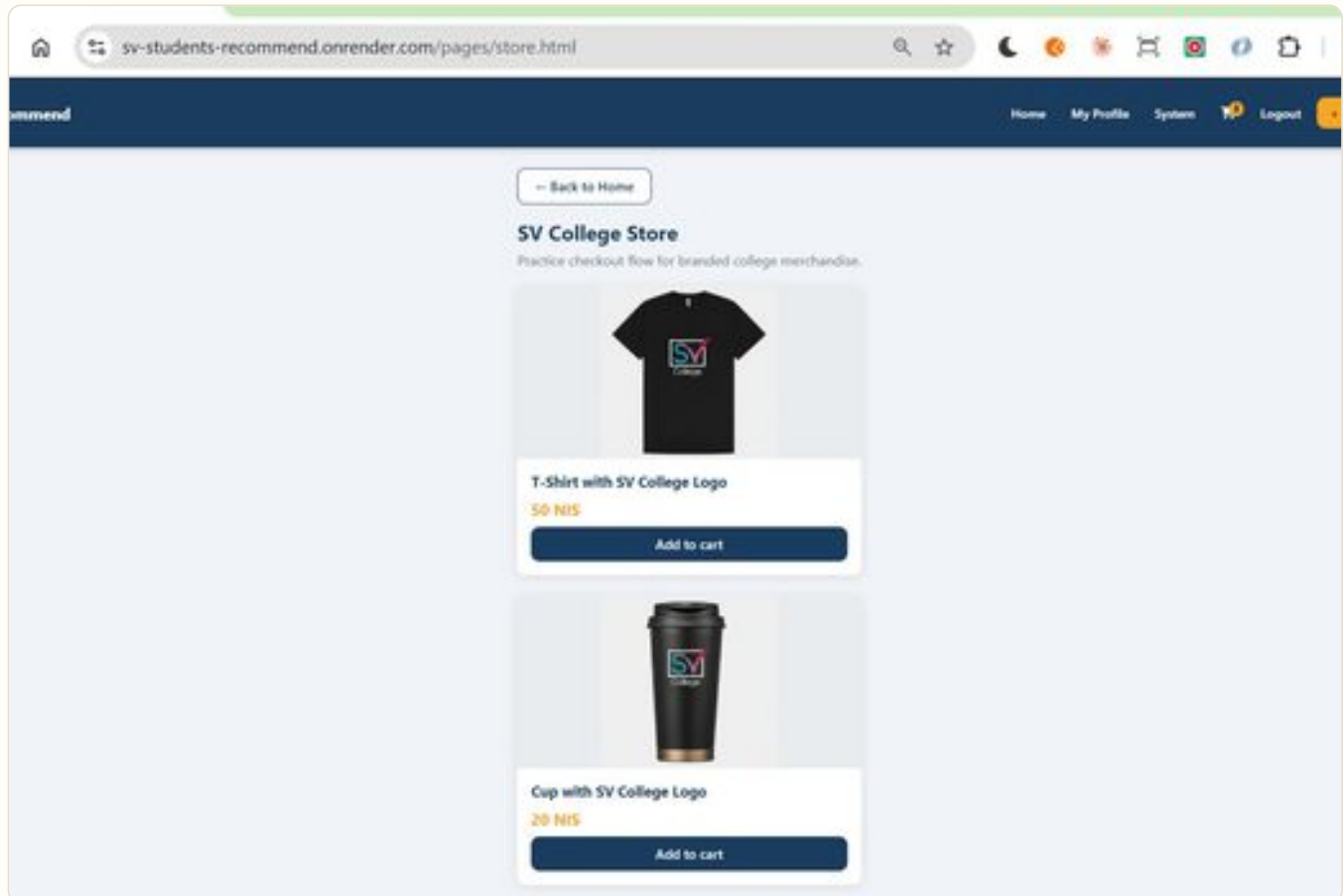
Website Link
https://...

Image

4.6 Store

URL <https://sv-students-recommend.onrender.com/pages/store.html>

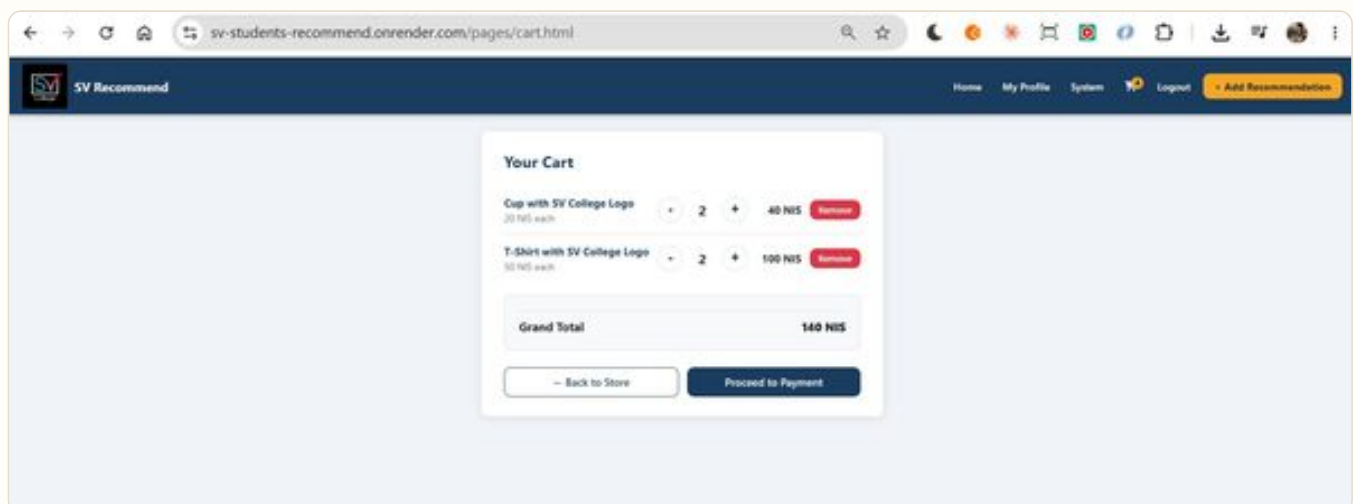
T-Shirt 50 NIS, Cup 20 NIS, Add to cart.



4.7 Cart

URL <https://sv-students-recommend.onrender.com/pages/cart.html>

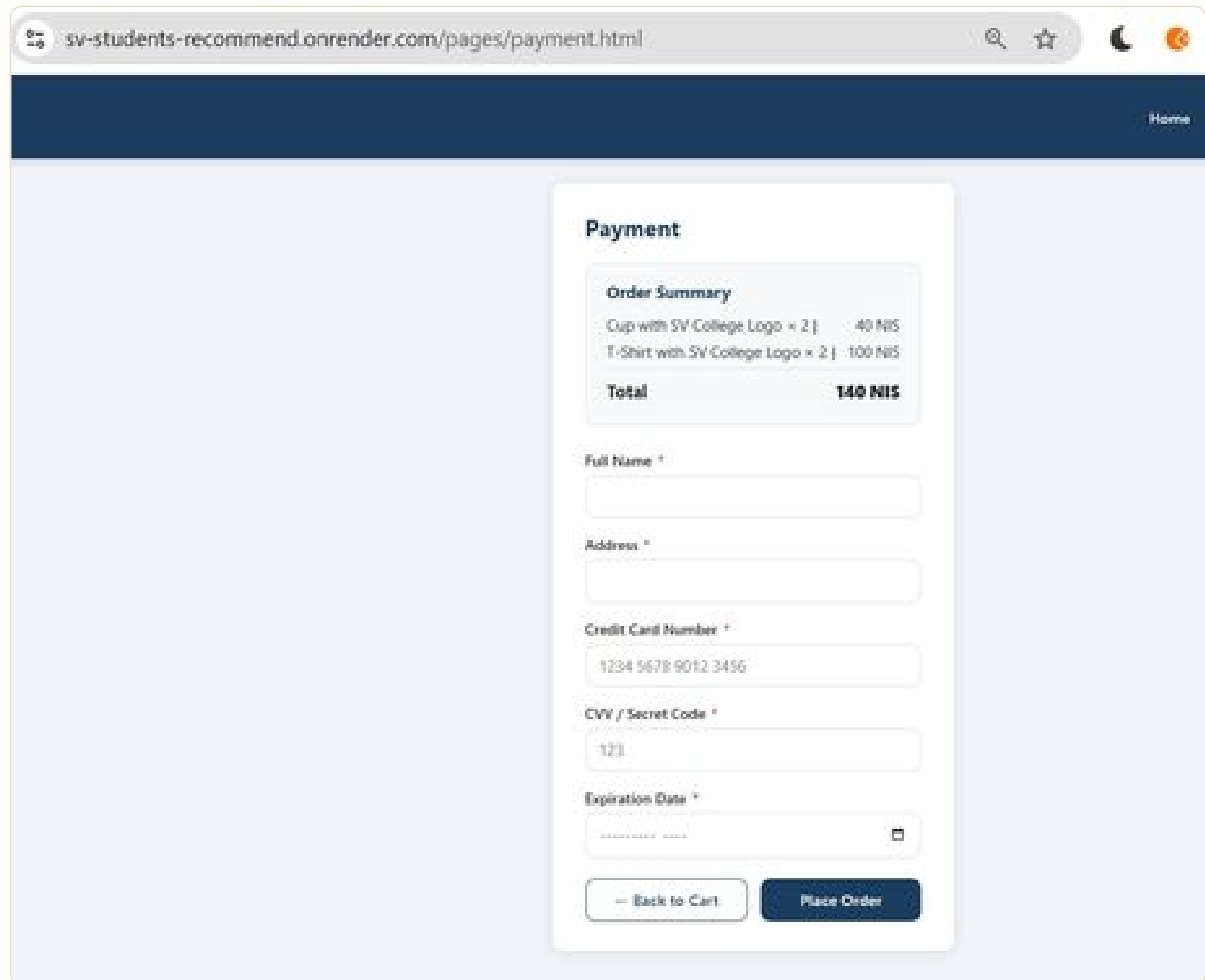
Change quantity, recalculated total, remove, proceed to payment.



4.8 Payment

URL <https://sv-students-recommend.onrender.com/pages/payment.html>

Order summary + mandatory fields, then Place Order.



The screenshot shows a web browser window with the URL sv-students-recommend.onrender.com/pages/payment.html. The page has a dark blue header with a "Home" link. The main content area is light blue and contains a white "Payment" form. The form includes an "Order Summary" section with the following items:

Order Summary	
Cup with SV College Logo x 2	40 NIS
T-Shirt with SV College Logo x 2	100 NIS
Total	140 NIS

Below the summary are five input fields with asterisks indicating they are required:

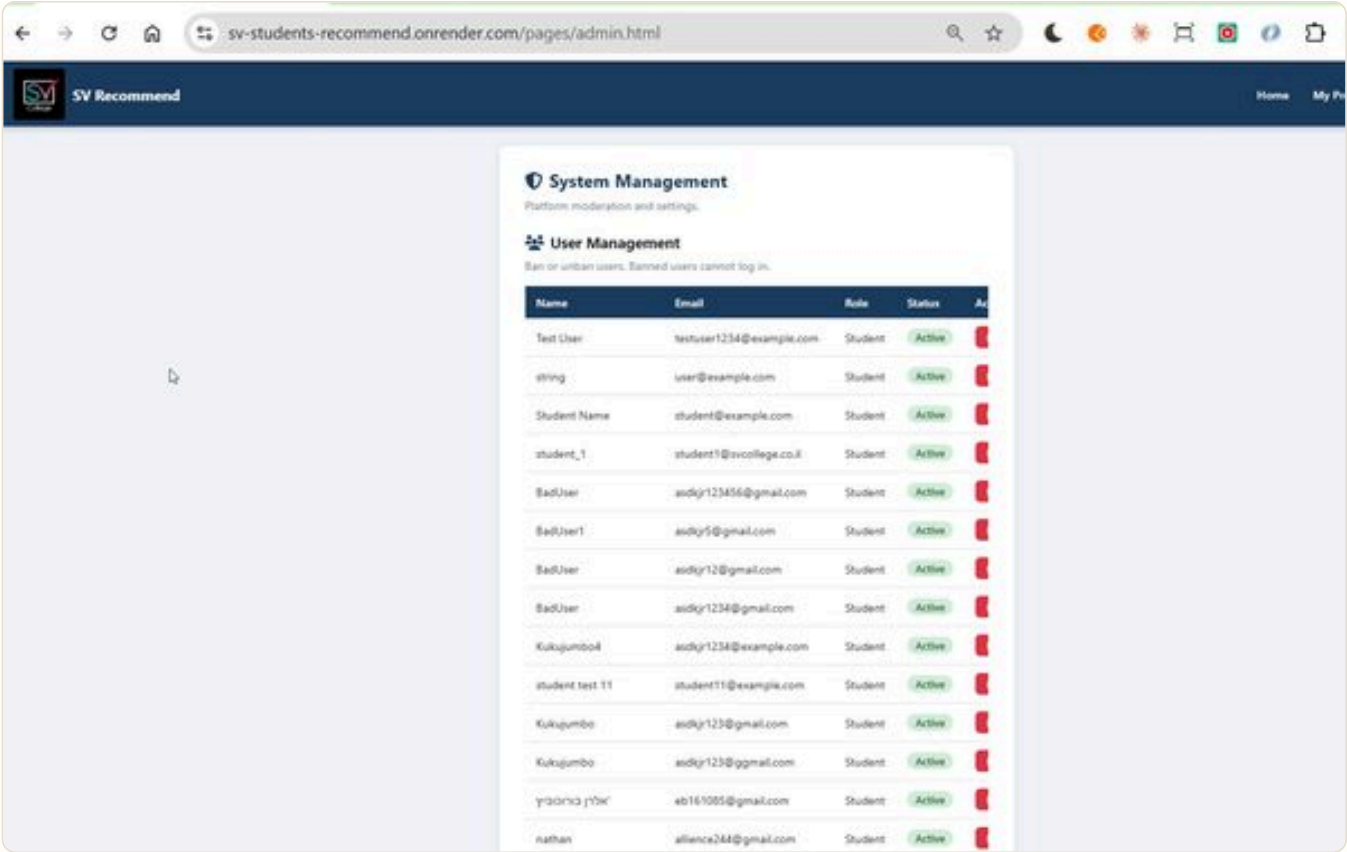
- Full Name *
- Address *
- Credit Card Number *
- CVV / Secret Code *
- Expiration Date *

The Credit Card Number field contains the value "1234 5678 9012 3456". The CVV / Secret Code field contains the value "123". The Expiration Date field is empty. At the bottom of the form are two buttons: "Back to Cart" and "Place Order".

4.9 System — User Management

URL <https://sv-students-recommend.onrender.com/pages/admin.html>

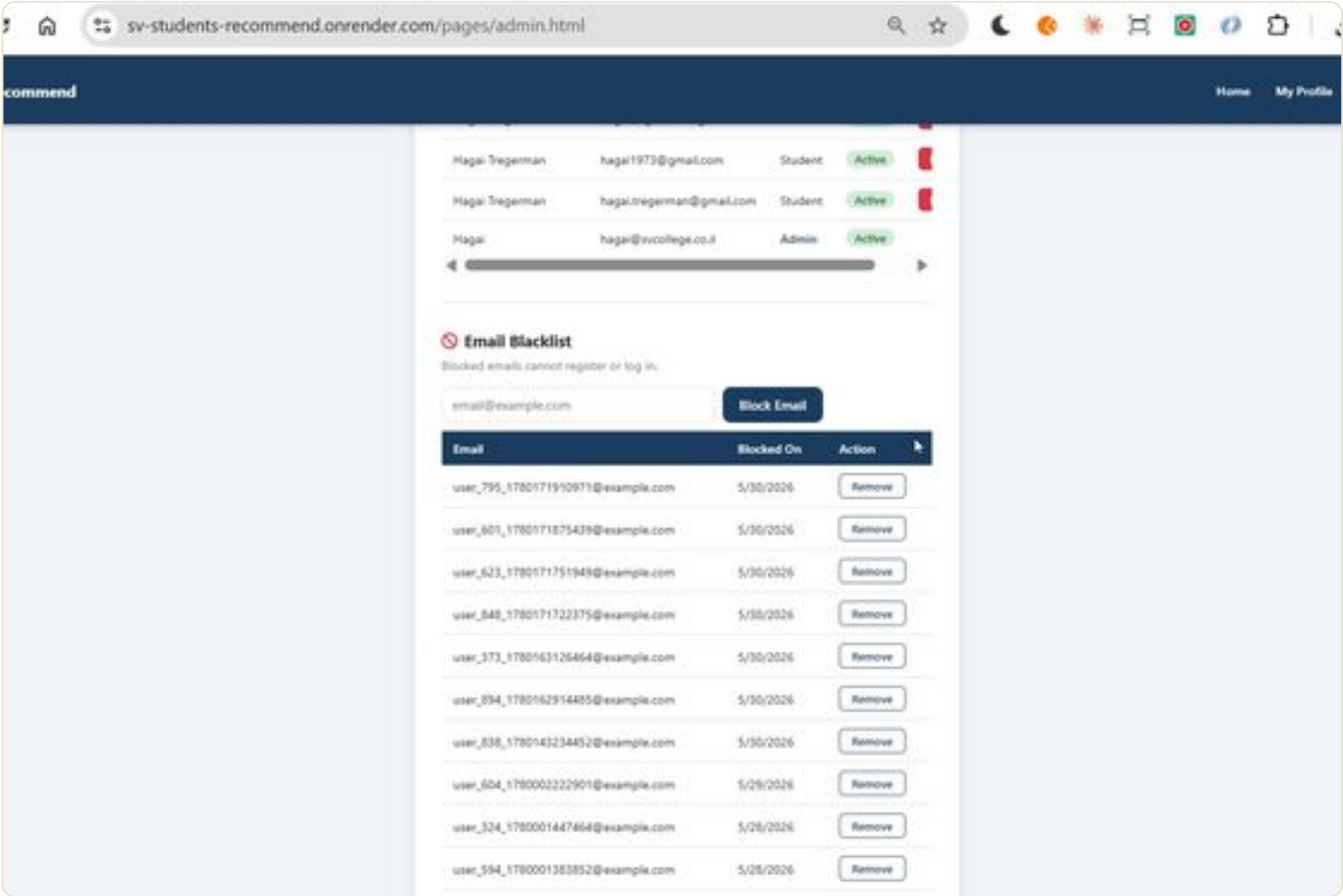
Ban / unban users (admin only).



4.10 System — Blacklist & Controls

URL <https://sv-students-recommend.onrender.com/pages/admin.html>

Block emails, suspend new recs, delete by ID.



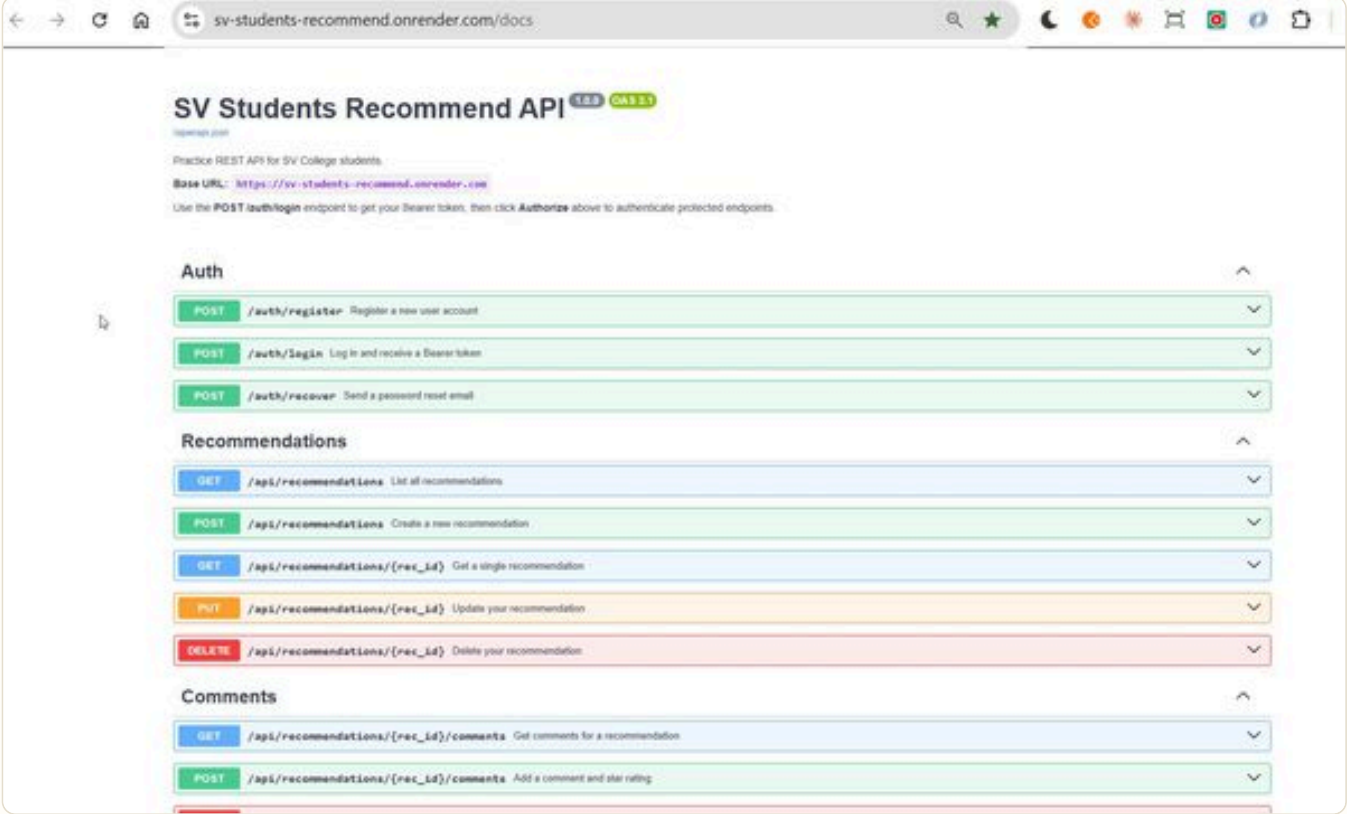
4.11 Footer

URL <https://sv-students-recommend.onrender.com/>
© 2026, SV College link, Accessibility link.



5. API Reference

Interactive Swagger docs: <https://sv-students-recommend.onrender.com/docs>. Base URL: `https://sv-students-recommend.onrender.com` . Get a token from `POST /auth/login` , then send `Authorization: Bearer <token>` on protected calls.



Method	Path	Auth	Purpose
POST	/auth/register	No	Register a new user.
POST	/auth/login	No	Log in, receive a token.
POST	/auth/recover	No	Send a reset email.
GET	/api/recommendations	No	List all recommendations.
POST	/api/recommendations	Yes	Create a recommendation.
GET	/api/recommendations/{rec_id}	No	Get one recommendation.
PUT	/api/recommendations/{rec_id}	Yes	Update your recommendation.
DELETE	/api/recommendations/{rec_id}	Yes	Delete your recommendation.
GET	/api/recommendations/{rec_id}/comments	No	Get comments.
POST	/api/recommendations/{rec_id}/comments	Yes	Add a comment + rating.

Status codes you will assert: 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found.

6. Store Your Project on GitHub

Your test project **must be stored in a GitHub repository** on github.com. This is how you submit your work and how it can be reviewed and run.

- 1 Create a free account at github.com if you don't have one.
- 2 Create a new repository, e.g. `sv-recommend-qa`. Add a `README.md` and a `.gitignore` for Python.
- 3 On your computer, put your Playwright tests in the repo folder using the structure in Section 7.
- 4 Commit and push:

```
# one-time setup inside your project folder
git init
git add .
git commit -m "Initial SRS test project"
git branch -M main
git remote add origin https://github.com/<your-username>/sv-recommend-qa.git
git push -u origin main
```

Good repo hygiene

Add a `venv/` line to `.gitignore` so you don't commit the virtual environment. Keep secrets (passwords) out of the code – read them from environment variables. Write a short README explaining how to install and run the tests.

7. Playwright Setup (Python)

```
# 1) virtual environment
python -m venv venv
# Windows: venv\Scripts\activate
# Mac/Linux: source venv/bin/activate

# 2) install
pip install pytest pytest-playwright requests
playwright install
```

7.1 Recommended structure

```
sv-recommend-qa/  
  conftest.py          # shared fixtures (base url, token)  
  test_api.py          # your chosen API tests  
  test_ui_positive.py  
  test_ui_negative.py  
  test_ui_boundary.py  
  requirements.txt  
  README.md
```

7.2 Record a test to find selectors

```
playwright codegen https://sv-students-recommend.onrender.com/pages/login.html
```

Use auto-waiting — never `sleep()`

Playwright waits until elements are ready. Instead of `time.sleep()`, assert the result:

```
expect(page).to_have_url("**/home.html") or expect(page.get_by_text("140 NIS")).to_be_visible() .
```

8. How to Write Each Type

8.1 API test

```
import time  
from playwright.sync_api import Playwright  
  
BASE = "https://sv-students-recommend.onrender.com"  
  
def test_register_new_user(playwright: Playwright):  
    api = playwright.request.new_context(base_url=BASE)  
    email = f"test_{int(time.time())}@example.com"  
    res = api.post("/auth/register", data={  
        "name": "Test Student", "email": email, "password": "abcd"})  
    assert res.status in (200, 201)  
    api.dispose()
```

8.2 Positive UI (E2E)

```
from playwright.sync_api import Page, expect  
  
def test_login_success(page: Page):  
    page.goto("https://sv-students-recommend.onrender.com/pages/login.html")
```

```
page.get_by_label("Email").fill("admin@svcollege.co.il")
page.get_by_label("Password").fill("YOUR_PASSWORD")
page.get_by_role("button", name="Sign In").click()
expect(page).to_have_url("**/home.html")
```

8.3 Negative / error handling

```
def test_login_invalid(page: Page):
    page.goto("https://sv-students-recommend.onrender.com/pages/login.html")
    page.get_by_label("Email").fill("nobody@example.com")
    page.get_by_label("Password").fill("wrongpass")
    page.get_by_role("button", name="Sign In").click()
    expect(page.get_by_text("Invalid", exact=False)).to_be_visible()
```

Golden rules

1. One test checks one thing. 2. Always assert. 3. Make it repeatable (unique email, clean up). 4. Name clearly, e.g. `test_password_exactly_6_chars_passes` . 5. No hard-coded waits.

Task A — API Test Bank

Choose and implement **6 of these 10** API tests. Tick the ones you pick; the counter tracks your selection.

API selected: 0 / 6 choose 6 more

Clear

Pick	Type	Test	Steps	Expected result	SRS
☐ A1	Positive	Register a new user	POST /auth/register with valid name, email, password.	200/201; user created.	3.1.2
☐ A2	Positive	Login success	POST /auth/login with correct credentials.	200; response contains a token.	3.1.1
☐ A3	Positive	Create recommendation (mandatory only)	POST /api/recommendations with only category, name, your name + valid token.	200/201; created.	3.3.3

Pick	Type	Test	Steps	Expected result	SRS
☐ A4	Negative	Create recommendation – empty category	POST /api/recommendations with category empty.	400 validation error; not created.	3.3.3
☐ A5	Positive (Admin)	Delete recommendation	Login as Admin, DELETE /api/recommendations/{rec_id} for an existing id.	200/204; deleted.	3.4.2
☐ A6	Negative (RBAC)	Regular user blocked from admin action	Login as regular user, attempt an admin-only / not-owned delete.	401 or 403; blocked.	2.1 / 2.2
☐ A7	Positive	List all recommendations	GET /api/recommendations (no auth needed).	200; returns a list/array.	3.3.1
☐ A8	Negative	Login with wrong password	POST /auth/login with a real email but wrong password.	401 / error; no token returned.	3.1.1
☐ A9	Positive	Add a comment + rating	Login, POST /api/recommendations/{rec_id}/comments with text and a 1–5 star rating.	200/201; comment created.	3.3.2
☐ A10	Negative	Create recommendation with no token	POST /api/recommendations without an Authorization header.	401 Unauthorized.	2.1

Task B — UI Test Bank

Choose and implement **14 of these 24** UI End-to-End tests. Try to cover all four groups.

UI selected: 0 / 14 choose 14 more

Clear

— Positive E2E Flows

Pick	Test	Steps	Expected result	SRS
☐ U1	Login flow	Open Login, enter valid email + password, click Sign In.	Redirected to Home page.	3.1.1

Pick	Test	Steps	Expected result	SRS
☐ U2	Store checkout flow	Store → add T-Shirt → Cart → Proceed to Payment → fill mandatory fields → Place Order.	Order placed / confirmation shown.	3.5.1-3.5.3
☐ U3	Create recommendation	+ Add Recommendation, fill all fields, submit.	New recommendation appears on Home.	3.3.3
☐ U4	Logo navigation	From any page click the SV College logo.	Returns to Home page.	3.2.1
☐ U5	Register then login	Register a brand new unique user, then log in with it.	Login succeeds; lands on Home.	3.1.2
☐ U6	Filter recommendations	On Home, click the “Movie” filter.	Only Movie cards are shown.	3.3.1
☐ U7	Add to cart updates counter	On Store, click Add to cart on one item.	Cart icon counter increases by 1.	3.5.1
☐ U8	Logout	While logged in, click Logout.	Session ends; redirected to Login page.	3.2.1 / 3.4.1

— Negative & Error-Handling

Pick	Test	Steps	Expected result	SRS
☐ U9	Invalid login	Unregistered email + random password, click Sign In.	Clear error shown; no redirect.	3.1.1
☐ U10	Payment validation – empty card	Leave Credit Card Number empty, click Place Order.	Validation error; submission blocked.	3.5.3
☐ U11	Add Recommendation – missing Your Name	Submit Add Recommendation without “Your Name”.	Form does not submit; error shown.	3.3.3
☐ U12	Access control via URL	Login as regular user, type /pages/admin.html in the address bar.	Access blocked / redirected away.	2.2 / 3.4.2
☐ U13	Banned user cannot log in	Admin bans a user; that user tries to log in.	Login denied with a message.	3.4.2
☐ U14	Blacklisted email cannot register	Admin blocks an email; try to register with it.	Registration rejected.	3.4.2
☐ U15	Payment – empty Full Name	Fill all payment fields except Full Name, Place Order.	Validation error; blocked.	3.5.3

Pick	Test	Steps	Expected result	SRS
☐ U16	Edit someone else's recommendation	As a regular user, open another user's recommendation.	No Edit/Delete available for it.	3.3.2

— Boundary Value

Pick	Test	Steps	Expected result	SRS
☐ U17	Password min – valid (6 chars)	Register with a password of exactly 6 English characters.	Registration succeeds.	3.1.2
☐ U18	Password min – invalid (3 chars)	Register with a password of exactly 3 characters.	Rejected with an error.	3.1.2
☐ U19	Cart quantity 0	In the cart set an item quantity to 0.	Item removed OR error; never negative.	3.5.2
☐ U20	Empty cart checkout	With an empty cart, try to proceed to payment.	Blocked or empty-cart message.	3.5.2

— Specific UI / Feature

Pick	Test	Steps	Expected result	SRS
☐ U21	Password show/hide	On Login, type a password (dots), click the eye icon.	Real text becomes visible; click again hides it.	3.1.1
☐ U22	Admin delete comment	Login as Admin, open a recommendation, Delete a comment.	Comment disappears.	3.3.2 / 2.2
☐ U23	Cart math recalculation	Add a Cup (20 NIS), set quantity to 2.	Line total and Grand Total update to 40 NIS.	3.5.2
☐ U24	Suspend new recommendations	Admin suspends new recs; a user tries + Add Recommendation.	Creation blocked while suspended.	3.4.2

What to Submit

- A **GitHub repository** link containing your test code (structured as in Section 7).
- Your 6 chosen API tests and 14 chosen UI tests, each with a clear name and assertions.
- A short results table: test name, pass/fail, and a note for any failure (it may be a real bug!).

- For any failed test: a screenshot or the error message.

Bonus challenge

Generate a fresh unique user every run. Add the blacklist test (admin blocks an email → it cannot register) and the suspend test (admin freezes new recommendations → a user cannot create one).

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